Executive Summary

Student Performance Analysis | Exploratory Data Analysis with Python

Project Objective

The project explores a dataset of about 30,600 students to understand which background factors are linked to performance in Math, Reading and Writing. It looks at gender, ethnic group, parental education, parental marital status and sport participation. It also checks score distributions for outliers. The aim is to find which factors matter and which don't, so support can be targeted where it helps most.

Skills Used

- **Programming and tools:** Python in a Jupyter Notebook, using Pandas and NumPy for data handling and Matplotlib and Seaborn for visualization.
- **Data loading and inspection:** reading CSV files, previewing data with head(), describe() and info(), and checking data types and missing values with isnull().
- **Data cleaning:** dropping an unneeded index column and identifying missing values across columns.
- **Data aggregation:** grouping with groupby() and agg() to compute average scores by category, and filtering with .loc to count group sizes.
- **Data visualization:** choosing suitable chart types (count plot, heatmap, box plot, pie chart), with labels, titles, color palettes and percentage annotations.
- **Statistical analysis:** descriptive statistics, comparison of group means, and outlier detection using box plots.
- **Analytical thinking:** drawing conclusions from charts, separating strong factors from negligible ones, and turning findings into recommendations.
- **Communication:** documenting the work with explanatory notes beside each chart.

Visualizations Used

- **Count plot:** gender distribution.
- **Heatmaps:** average scores by parental education, parental marital status and sport practice.
- **Box plots:** outlier checks on Math, Reading and Writing scores.
- **Pie chart:** share of students by ethnic group.

Key Insights

1 **Parental education shows the strongest link to scores.** Students whose parents hold a master's degree average about 72 in Math, 76 in Reading and 76 in Writing. Those whose parents attended only some high school average about 63, 66 and 64, a gap of roughly 10 to 13 points. Scores rise steadily with each education level.

2. **Parental marital status has almost no effect.** Average scores across divorced, married, single and widowed households differ by about 1 point or less.
3. **Regular sport practice is linked to somewhat higher scores.** Students who practice regularly lead those who never do by about 3.7 points in Math, 3.1 in Writing and 1.6 in Reading. The gap is modest, and "sometimes" falls in between.
4. **Gender split is nearly even.** The data has 15,424 females and 15,217 males, so the female majority is only about 0.7%.
5. **Ethnic group composition is uneven.** Group C is the largest (32.0%), followed by D (26.1%), B (20.2%), E (14.0%) and A (7.7%).
6. **Outliers sit only at the low end.** All three subjects have clusters of very low scores, and Math includes scores of 0. There are no high-end outliers, since scores are capped at 100.
7. **Data quality needs attention.** Several columns have missing values, such as transport (about 10%), ethnic group, test prep and parental education. The notebook does not yet handle these.

Business Recommendations

1. **Target support at students from lower parental-education backgrounds.** Tutoring, mentoring, homework help and study resources would address the largest performance gap.
2. **Engage parents.** Offer parent workshops and guidance on supporting learning at home, especially for families with less formal education.
3. **Encourage regular sport and extracurricular participation.** The link is modest and doesn't prove causation, but it is low-cost and has other benefits.
4. **Don't use marital status as a basis for interventions.** The data shows no meaningful impact.
5. **Identify and support low-scoring students early.** Review the low-end outliers to find those who need remedial help, and check for data-entry errors such as the zero scores.
6. **Improve data quality.** Handle missing values before further modeling.
7. **Extend the analysis.** Weekly study hours, test preparation, lunch type and ethnic group versus scores are in the dataset but not yet analyzed. Statistical tests or regression would show which factors matter most once the others are controlled for.

Note: these findings are correlations from averages, not proof of cause and effect.